

MakeCode Help Desk Starter Blocks

NAME _____ DATE _____ PERIOD _____

This is the starter program, block by block, in the order you drag them in. Build it at makecode.microbit.org, then change only the words inside the quote marks to your team's three troubleshooting steps. There is no link to click and no file to download. You build it from this card.

Before you start: from the **Variables** drawer, click **Make a Variable** and name it `step`.

#	DRAWER	BLOCK TO DRAG	WHERE IT GOES	WHAT TO TYPE IN IT
1	Variables	set step to 0	inside on start	change the 0 to 1
2	Basic	show string " "	inside on start, under block 1	Help Desk Ready
3	Input	on button A pressed	empty workspace	nothing
4	Logic	if / then / else	inside on button A pressed	nothing yet
5	Logic	0 = 0 comparison	into the if slot of block 4	drag step into the left side, type 1 on the right
6	Basic	show string " "	inside the first then	Step 1: plus your first step
7	Logic	click the + on the if block	adds an else if	nothing
8	Logic	0 = 0 comparison	into the else if slot	drag step into the left side, type 2 on the right
9	Basic	show string " "	inside the else if then	Step 2: plus your second step
10	Basic	show string " "	inside the final else	Step 3: plus your third step
11	Variables	change step by 1	inside on button A pressed, under the whole if block	nothing
12	Logic	if / then	under block 11	nothing
13	Logic	0 > 0 comparison	into the if slot of block 12	drag step into the left side, type 3 on the right
14	Variables	set step to 0	inside that then	change the 0 to 1

#	DRAWER	BLOCK TO DRAG	WHERE IT GOES	WHAT TO TYPE IN IT
15	Input	on button B pressed	empty workspace	nothing
16	Basic	show string " "	inside on button B pressed	FIXED!

Read the order carefully. Block 11 comes AFTER the whole if block, not before it. The micro:bit shows the step first, then counts up to the next one. If you put `change step by 1` above the if block, the first press of Button A skips straight to Step 2.

Test it. Press A four times. You should see Step 1, Step 2, Step 3, and then Step 1 again. Press B. You should see FIXED!

Teacher, rebuilding this in about 60 seconds. Make the variable, drop `set step to 1` and `show string "Help Desk Ready"` in `on start`, add `on button A pressed` with a three-branch if on `step` and one `show string` per branch, put `change step by 1` under the if with a `if step > 3 then set step to 1` guard beneath it, then `on button B pressed` with `show string "FIXED!"`. Build it once on the projector before class and leave the tab open. Any student who needs it can copy it off the screen.